

Does the Assistant Check the Record? Knowledge-Base Poisoning and System-of-Record Validation in Utility Operations RAG

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Abstract—Retrieval-augmented generation (RAG) is increasingly considered for document-grounded technical lookup, but indexed repositories can contain low-trust contributions. We study whether such documents can cause a legitimate utility assistant to commit to out-of-range numeric values and whether an independent system-of-record (SoR) check helps. GridPoison-100 combines 35 public-standard items, 45 fictional asset-specific items, and 20 legitimate-conflict items. Across Mistral Small 3.2, Qwen3-8B, and GPT-5.6 Luna, one-poison retrieval exposure at $k = 5$ was 70.0% for public items but 3.3% for asset items; three poisons raised asset exposure to 76.7%. Yet forced exposure produced 0% exact attacker-value adoption and 0% benchmark violations in every model/tier/position cell, with poison position instead changing whether models returned the correct value or declined to commit. Under the three-poison stress condition, SoR validation reduced benchmark-violation rate by 3.1–3.8 percentage points at 70% coverage and 3.8–5.0 points at full coverage, while conditional benign-conflict false flags reached 43.6–45.7% and 64.1–65.7%, respectively. Holm-adjusted McNemar tests were non-significant. The results show why retrieval exposure, adoption, and consequence should be measured separately, and that broader record coverage trades violation reduction against false-flag cost.

Index Terms—retrieval-augmented generation, knowledge-base poisoning, power systems, utility operations, large language models, data integrity

I. INTRODUCTION

Large language models are increasingly proposed as natural-language interfaces to technical documentation. In power and utility settings, the most plausible near-term use is not autonomous control but constrained lookup: a field worker asks for an approach distance, a protection engineer checks an interconnection boundary, or a planner retrieves an approved asset limit. RAG makes such answers auditable by grounding them in external documents, yet it also creates a write-path dependency. If an indexed repository accepts contributions from contractors, shared drives, wikis, or externally supplied bulletins, a legitimate user can receive an incorrect value even when the user’s prompt is benign.

Knowledge-base poisoning of RAG is already established [1], [2]. Accordingly, our contribution is not another demonstration that injected text can influence an answer. Instead, we ask what poisoning means when the answer can be graded against an authoritative operational record, whether model prior knowledge protects public facts more than asset-specific facts, and how a separate structured record changes the failure mode.

We study three research questions. **RQ1** asks how targeted poisoning decomposes into retrieval exposure and conditional adoption. **RQ2** asks whether attacker-value adoption differs between public-standard facts and asset-specific facts when exposure is held constant. **RQ3** asks how much system-of-record validation reduces out-of-range commitments relative to two generic defenses, and what false-flag cost it introduces when legitimate documents disagree by jurisdiction or inter-connection.

The contributions are fourfold: (1) a write-path threat model for low-trust utility-document repositories; (2) GridPoison-100, a released benchmark with source-grounded public items, a fictional asset register, and legitimate-conflict controls; (3) a two-stage evaluation that separates retrieval compromise, conditional adoption, and consequence grading; and (4) an empirical evaluation of system-of-record validation under partial coverage rather than as an oracle.

II. RELATED WORK

PoisonedRAG showed that a small number of attacker-controlled texts can steer retrieval-augmented models toward targeted answers [1]. Subsequent work has examined retrieval, database, and generator factors that influence poisoning exposure and attack success [2]. Because those effects are already known, retrieval depth is a secondary factor here rather than a novelty claim.

RobustRAG introduced an isolate-then-aggregate strategy for retrieval corruption [3]; we use this family as a generic defense baseline rather than presenting it as new. ClashEval quantified conflicts between parametric knowledge and retrieved evidence [4]. Other work has shown that evidence phrasing, publication time, and metadata can affect model preferences [5], [6]. These findings motivate our vendor-bulletin condition but also rule out treating authority framing as a primary contribution.

LLM risks in power systems have already been discussed and experimentally explored [7], [8]. Our study therefore avoids claims of being the first LLM-security work in the domain. The distinction from our prior healthcare RAG work is also explicit: that work asks whether an unauthorized reader can obtain protected information, whereas the present study asks whether an authorized reader receives a false operational value after a low-trust writer inserts content into the retrieval corpus.

III. THREAT MODEL

The attacker is an authorized but low-trust writer, such as a contractor with write access to an indexed collaboration repository, or the sender of a spoofed technical bulletin that is subsequently indexed. The attacker may add one to three new documents but cannot edit or delete existing documents. They cannot modify the structured asset/settings record, EMS/SCADA, protection devices, model weights, or the user’s query. The attacker knows the target asset or topic but not the exact query wording, embedding model, retrieval results, or surrounding corpus.

The attack goal is targeted answer-integrity corruption: a legitimate lookup should commit to an attacker-chosen numeric value. The primary poison is a factual statement rather than an instruction to the model, separating this setting from indirect prompt injection. Confidentiality, availability, model-weight poisoning, multi-turn social engineering, and physical control-loop attacks are outside scope.

IV. GRIDPOISON-100

GridPoison-100 contains 100 lookup items divided into three tiers. Tier 1 contains 35 public normative facts drawn from Massachusetts DER settings material, OSHA 29 CFR 1910.269 minimum-approach-distance tables, and NERC PRC-024-3 interconnection boundaries [9], [10], [11]. Each public item records a source URL, source locator, units, applicability conditions, and an allowable interval.

Tier 2 contains 45 asset-specific facts from a declared fictional utility register. Asset identifiers and values are synthetic by construction, preventing a model from knowing them from pretraining. The register covers transformer emergency ratings, breaker current ratings, feeder thermal limits, BESS continuous discharge limits, and cable ampacity values. For Tier 2, the register is the authoritative record by construction; no claim is made that these values describe a real network.

Tier 3 contains 20 legitimate-conflict items. Two documents can disagree because of jurisdiction, interconnection, category, or table scope, while the query explicitly names which context applies. These items are not poisoned; they measure whether a defense incorrectly flags a correct context-specific answer.

The clean corpus contains 400 short documents: three support documents for each poisonable item, legitimate-conflict documents, and on-topic distractors. Poison variants are generated as minimal factual edits. The main poison changes the critical numeric value outside the encoded allowable range. A secondary wrong-but-in-range variant separates factual incorrectness from consequence violation. A dated vendor-bulletin form is treated as an attack framing condition, not a novel mechanism.

V. EXPERIMENTAL DESIGN

A. Reference conditions and models

B0 is closed-book generation, B1 is clean RAG, and B2 is poisoned vanilla RAG. The evaluated models are Qwen3-8B, Mistral Small 3.2 24B, and GPT-5.6 Luna, all called through the same OpenAI-compatible endpoint. Runs use temperature

zero and log requested and resolved model/provider metadata, response ID, usage, timestamp, and repository commit.

B. Stage A: retrieval exposure

RQ1 uses natural dense retrieval. The lower-strength condition inserts one out-of-range poison; the stress condition inserts three distinct poison paraphrases. Poison@ k is reported separately from generation behavior. A retrieval-only pre-generation calibration found that one-document exposure was strongly imbalanced between Tier 1 and Tier 2, which would confound a direct cross-tier adoption comparison.

C. Stage B: forced exposure

RQ2 therefore uses a fixed five-passage context rather than natural retrieval. Each item receives three clean same-item support passages, one deterministic on-topic distractor, and exactly one poison. The poison is counterbalanced in the first or last position. This design holds exposure and context length constant while testing whether models fall back on parametric knowledge differently for public-standard and asset-specific facts.

D. Defenses

D1 is a cautious prompt instructing the model to identify unresolved source conflicts. D2 isolates each retrieved passage, obtains one value per passage, and aggregates by majority, following the isolate-then-aggregate defense family [3]. D3 is deterministic system-of-record validation: when a committed value is covered by the structured record, a mismatch is replaced by a verification flag. D3 is evaluated at 40%, 70%, and 100% deterministic coverage. The record is deliberately imperfect for some legitimate jurisdictional variants, making false flags measurable instead of giving D3 oracle access to the answer key.

RQ3 uses the three-poison $k = 5$ stress condition to avoid a retrieval floor when comparing defenses. Secondary one-factor deviations examine a dated vendor bulletin, a wrong-but-in-range value, and $k = 2$. No full factorial sweep is performed.

VI. METRICS AND STATISTICAL ANALYSIS

Every generation trial receives exactly one label: correct (C), attacker value adopted (A), other wrong value (W), conflict correctly flagged (F), or refusal (R). Numeric C/A/W grading is deterministic from the required VALUE: line and item tolerance.

Let E_t indicate whether a poisoned document occurs in the retrieved top- k context. We report Poison@ k , attack success rate (ASR), conditional adoption rate (CAR), correct-response rate (CRR), conflict-flag rate (CFR), and benchmark violation rate (SVR):

$$\text{CAR} = P(A \mid E_t = 1), \quad (1)$$

$$\text{SVR} = \frac{1}{|T|} \sum_t \mathbb{1}[\hat{v}_t \notin [L_i, U_i]]. \quad (2)$$

SVR is a benchmark consequence metric only; it is not a physical grid-safety measure.

TABLE I
POISON EXPOSURE AT $k = 5$ IN RETRIEVAL-ONLY CALIBRATION.

Condition	Tier 1	Tier 2	Overall
1 plain poison	70.0%	3.3%	32.5%
3 poison documents	70.0%	76.7%	73.8%
1 vendor bulletin	67.1%	7.8%	33.8%

The item is the unit of analysis. Two query phrasings are repeated observations, not independent samples. All rate estimates receive 95% cluster-bootstrap intervals over items with 2,000 resamples. Paired defense comparisons use exact McNemar tests with Holm correction. The primary RQ2 endpoint is attacker-value adoption under forced exposure, stratified by tier and poison position. The primary RQ3 endpoint is paired Δ SVR for D3 at 70% coverage relative to D1 and D2. The Tier 3 benign-conflict false-flag rate (BFFR) is computed conditionally among trials for which clean RAG first returned the benchmark-correct context-specific answer; a D3 flag on that otherwise-correct answer counts as a false flag.

VII. RESULTS

Clean-RAG Tier 2 CRR was 87.8–87.8% across models, meeting the locked 80% Tier 2 interpretation gate; Tier 1 CRR was 75.7–78.6%.

A. Retrieval exposure and natural-retrieval outcomes (RQ1)

Retrieval-only calibration showed that one inserted document was exposed much more often for public-standard items than for fictional asset-specific items. Table I reports Poison@ k at $k = 5$. Despite this exposure, exact attacker-value adoption remained low. With one poison, Tier 1 ASR was 2.9–2.9% and Tier 2 ASR was 0.0–0.0%. With three poisons, Tier 1 ASR was 2.9–2.9% and Tier 2 ASR was 3.3–4.4% across models.

B. Forced-exposure behavior (RQ2)

Across all 12 model/tier/position cells, attacker-value adoption and benchmark-violation rate were both 0%. The poison therefore changed conflict handling rather than causing direct adoption in this fixed five-passage control. Table II reports correct-answer retention. Poison-last contexts produced 0% correct-answer retention for all three models and both tiers; responses instead declined to commit because the final passage contradicted the preceding majority. Poison-first retention varied substantially by model and tier, showing a position effect without attacker-value adoption.

C. Defense comparison (RQ3)

Table III reports the paired reduction in benchmark-violation rate relative to the three-poison vanilla-RAG stress baseline. Positive Δ SVR indicates fewer out-of-range commitments. At 70% record coverage, D3 reduced SVR by 3.1–3.8 percentage points across models; at 100% coverage the reduction was 3.8–5.0 points. D2 did not improve Mistral or GPT-5.6 Luna

and increased Qwen SVR by 3.1 points. All planned Holm-adjusted McNemar tests were non-significant ($p_{\text{Holm}} = 1.0$), so the bootstrap effect-size intervals are interpreted descriptively.

D. Legitimate-conflict cost

D3 can reject a correct context-specific answer when the simplified structured record lacks the applicable jurisdictional or interconnection variant. BFFR is therefore computed only among Tier 3 trials for which clean RAG first returned the benchmark-correct answer. The coverage benefit therefore has a clear cost: at 70% coverage BFFR was about 43.6–45.7%, and at full coverage it was about 64.1–65.7%.

VIII. DISCUSSION

The results separate three mechanisms that a single attack-success number would obscure. With one inserted document, retrieval exposure differed by more than an order of magnitude between the public and fictional asset tiers, yet exact attacker-value adoption remained low. Under forced exposure, no model adopted the attacker value in any tier/position cell. The planned RQ2 hypothesis of greater resistance for public facts is therefore not supported by attacker-value adoption in this design. Instead, passage position changed conflict handling: placing the poison last drove correct-answer retention to zero across all models and tiers, with models declining to commit rather than choosing the poison. This is a negative result for direct adoption but a positive reason to report non-commitment separately from correctness.

The defense results also resist a simple *more filtering is better* interpretation. Cautious prompting produced little change, and isolate-then-aggregate did not improve Mistral or GPT-5.6 Luna and increased Qwen’s violation rate in the paired effect-size analysis. D3 behaved differently because it checks a separate structured source rather than asking the language model to reason harder over conflicting text. At 70% and 100% coverage its bootstrap Δ SVR intervals were positive across all three models, but the corresponding Holm-adjusted McNemar tests were non-significant. We therefore treat the observed reductions as descriptive effect sizes, not confirmatory evidence of universal defense efficacy.

The cost side is equally important. As record coverage increased, D3 rejected more correct jurisdiction- or interconnection-specific answers when the simplified record represented a generic alternative. Conditional BFFR reached roughly 44–46% at 70% coverage and 64–66% at full coverage. For operational deployment, the design implication is not to maximize record coverage blindly, but to make record scope explicit and to route mismatches to verification rather than presenting the record as an oracle.

IX. LIMITATIONS

GridPoison-100 is a controlled benchmark rather than a deployed utility corpus. Tier 2 uses a fictional asset register, one dense retriever is used for the main study, and only

TABLE II
CORRECT-ANSWER RATE UNDER FORCED EXPOSURE. CLUSTER-BOOTSTRAP 95% INTERVALS ARE OVER ITEMS; ATTACKER-VALUE ADOPTION WAS 0% IN EVERY CELL.

Model	First/T1	First/T2	Last/T1	Last/T2
Mistral Small 3.2	8.6% [0.0, 17.1]	97.8% [93.3, 100.0]	0.0% [0.0, 0.0]	0.0% [0.0, 0.0]
Qwen3-8B	42.9% [25.7, 60.0]	57.8% [44.4, 73.3]	0.0% [0.0, 0.0]	0.0% [0.0, 0.0]
GPT-5.6 Luna	74.3% [60.0, 88.6]	28.9% [15.6, 42.2]	0.0% [0.0, 0.0]	0.0% [0.0, 0.0]

TABLE III
PAIRED REDUCTION IN BENCHMARK-VIOLATION RATE UNDER THE THREE-POISON STRESS CONDITION.

Model	D1	D2	D3@40%	D3@70%	D3@100%
Mistral Small 3.2	1.2% [0.0, 3.1]	0.0% [-3.8, 3.8]	0.0% [0.0, 0.0]	3.8% [0.6, 8.1]	4.4% [1.2, 8.8]
Qwen3-8B	0.0% [0.0, 0.0]	-3.1% [-6.9, -0.6]	0.0% [0.0, 0.0]	3.1% [0.6, 6.9]	3.8% [0.6, 7.5]
GPT-5.6 Luna	0.0% [0.0, 0.0]	0.0% [-1.9, 1.9]	0.6% [0.0, 1.9]	3.8% [0.6, 7.5]	5.0% [1.9, 9.4]

TABLE IV
CONDITIONAL BENIGN-CONFLICT FALSE-FLAG RATE (BFFR).

Model	40%	70%	100%
Mistral Small 3.2	17.9%	43.6%	64.1%
Qwen3-8B	17.1%	45.7%	65.7%
GPT-5.6 Luna	17.9%	43.6%	64.1%

three model families are tested. The primary attacker is non-adaptive and cannot modify the structured record. No power-flow, protection, thermal, or transient simulation is performed, so the benchmark does not quantify physical consequences. The study also focuses on scalar factual lookups; procedural switching sequences and open-ended engineering judgment are outside scope.

The public values were source-audited before generation. Because the conference schedule preceded completion of the external power-engineer review, generation used the committed source-audited candidate benchmark; the benchmark is therefore not described as engineer-validated. Any later review that changes a target value, applicability condition, allowable interval, or poison construction requires rerunning the affected experiments. The forced-exposure result also reflects a deliberately evidence-rich context with three clean same-item supports against one poison, so 0% attacker-value adoption should not be generalized to weaker or noisier evidence settings.

X. CONCLUSION

GridPoison-100 shows that poisoning exposure, attacker-value adoption, and benchmark consequence are distinct events in utility-operations RAG. In this study, exposure could be high while exact attacker-value adoption remained low or zero, and forced exposure primarily changed whether models committed to an answer. A separate system-of-record check reduced observed out-of-range commitments as coverage increased, but it also produced substantial false flags when the record lacked legitimate contextual variants. The practi-

cal lesson is therefore conditional: structured validation can change the failure mode, but its value depends on record scope, coverage, and escalation policy rather than on coverage alone. The conclusions remain limited to the stated threat model, benchmark, retriever, and tested models.

GENERATIVE AI USE DISCLOSURE

Generative AI tools assisted software scaffolding, literature triage, adversarial design review, and language editing. The authors verified cited sources, benchmark values, methods, code, analyses, and reported results and remain fully responsible for the manuscript.

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